Imaginary-Time Evolution in Quantum Systems: Extracting Ground and Excited States through Wick Rotation

Syed Hussain Haider, Qiongyuan Wu, James Millen syed.h.haider@kcl.ac.uk

Department of Physics, King's College London, Strand, London WC2R 2LS UK

SCHRÖDINGER–HEAT ANALOGY

The time-dependent Schrödinger equation

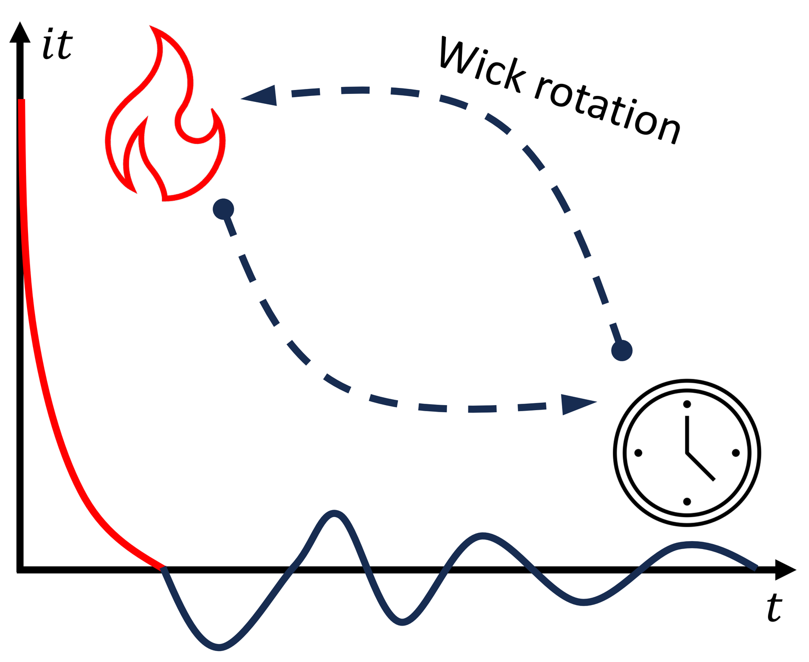
$$i\hbar\frac{d}{dt}|\Psi\rangle = \hat{H}|\Psi\rangle \quad (1)$$

has a mathematical form similar to the classical heat equation

$$\frac{\partial\Phi}{\partial t} = K\nabla^2\Phi \quad (2)$$

The only difference is the presence of the imaginary unit i in front of the time derivative. This simple distinction changes everything: while the heat equation produces diffusive decay, the Schrödinger equation produces oscillations. For a heat mode e^{-Dk^2t}, amplitude decreases with time, whereas for a quantum mode $e^{-i\omega t}$, amplitude oscillates.

Under a Wick rotation $(t \to -i\tau)$, the Schrödinger equation transforms into a diffusion equation, potentially with an additional potential term arising from the Hamiltonian.. This observation suggests a deep connection between diffusion and quantum evolution, which serves as the starting point for further exploration in this work.



IMAGINARY-TIME EVOLUTION

Applying the Wick rotation to the Schrödinger equation gives

$$\frac{\partial\psi(x,\tau)}{\partial\tau} = -\frac{\hat{H}}{\hbar}\psi(x,\tau) \quad (3)$$

For a harmonic oscillator Hamiltonian

$$\hat{H} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2 \quad (4)$$

the solution can be expanded in its stationary eigenbasis as

$$\psi(x,\tau) = \sum_n c_n\phi_n(x)e^{-\frac{E_n\tau}{\hbar}} \quad (5)$$

Factoring out the ground-state exponential gives

$$\psi(x,\tau) = e^{-\frac{E_0\tau}{\hbar}}\left[c_0\phi_0(x) + \sum_{n>0} c_n\,\phi_n(x)e^{-\frac{(E_n-E_0)\tau}{\hbar}}\right] \quad (6)$$

As $\tau \to \infty$, all higher–energy exponentials vanish, leaving only the lowest–energy state. Thus, in imaginary time, the wavefunction gradually relaxes toward the lowest–energy configuration of the system. This shows that diffusion-like evolution naturally suppresses higher–energy contributions, leaving behind the ground state as $\tau \to \infty$,

$$\psi(x,\tau\to\infty) \propto \phi_0(x) \quad (7)$$

NUMERICAL FRAMEWORK

In systems with known analytical solutions, the energy levels can be obtained directly by choosing $n = 0,1,2,\ldots$ in the Schrödinger equation. However, most real potentials lack exact solutions, making imaginary-time propagation a powerful computational approach.

To implement this, I constructed a finite-difference Hamiltonian on a one-dimensional grid and used the Crank–Nicolson scheme for imaginary-time evolution. The update at each step involves solving a tridiagonal system $A\psi^{(k+1)} = B\psi^{(k)}$ using a Thomas algorithm, where the matrices A and B encode the kinetic and potential energy operators. After each iteration, the wavefunction is normalised to maintain unit probability, and the energy is computed via the Rayleigh quotient,

$$E[\psi] = \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle}. \quad (8)$$

Convergence to the ground state is reached when the change in energy between iterations falls below a set tolerance.

Once the ground state $\phi_0(x)$ is obtained, the corresponding energy E_0 and wavefunction are stored, and deflation is performed to isolate higher states. The next trial wavefunction is orthogonalised against all previously found states using a Gram–Schmidt procedure:

$$\psi' = \psi - \sum_{i=0}^{m-1}(\langle\phi_i|\psi\rangle)\phi_i \quad (9)$$

The same imaginary–time iteration is then repeated on this residual to obtain the next eigenstate. Iterating this process successively yields multiple bound states, allowing reconstruction of the low–energy spectrum for arbitrary one–dimensional potentials (see code repository [3]).

RESULTS

The numerical solver was applied to three representative systems to test the effectiveness of the imaginary-time Crank–Nicolson framework: a quartic potential, a position-dependent mass Schrödinger equation (PDMSE), and the hydrogen atom, using a unified code framework available in [3].

Each potential required small adjustments to the spatial domain or boundary conditions, but the same numerical scheme was used in all cases.

For the quartic potential, $V(x) = x^2 + 0.1x^4,; x \in (-\infty,\infty)$, the computed energies reproduced the reference spectrum from the literature [1] with agreement up to four significant figures. This provided a first benchmark for validating the stability and accuracy of the method. The results are summarised in Table 1.

n	Computed E_n	Ref. E_n
0	1.065238	1.065286
1	3.306602	3.306872
2	5.747178	5.747959
3	8.351025	8.352678
4	11.095647	11.098600

Table 1: Quartic: $V(x) = x^2 + 0.1x^4$, $x \in (-\infty,\infty)$.

The PDMSE case was reformulated as a time-dependent Schrödinger equation with constant mass and modified effective potential following the approach described in [2]. Because this potential is defined on a finite spatial domain, the numerical grid and boundary conditions were adjusted accordingly. The computed energy eigenvalues, shown in Table 2, are consistent with the reference results for the same system.

n	Computed E_n	Ref. E_n
0	1.950409	1.950334
1	6.011908	6.011578
2	12.009202	12.008326
3	20.007291	20.005520
4	30.006896	30.003847

Table 2: PDMSE: $V(z) = \frac{1}{2} + \frac{3}{4}\tan^2 z + \sin z$, $z \in [-\pi/2, \pi/2]$.

For the hydrogen atom, the potential was softened to avoid the Coulomb singularity at the origin,

$$V_{\text{soft}}(r) = -1/\sqrt{(r^2+\varepsilon^2)} + \ell(\ell+1)/(2r^2),$$

with $\ell = 0$. This simplified form allowed stable propagation under imaginary time while retaining the expected $E_n \propto -1/(2n^2)$ pattern.

The computed energy levels are listed in Table 3 and compared against the analytical values.

n	Computed E_n	Exact E_n
0	-0.497461	-0.500000
1	-0.123492	-0.125000
2	-0.057041	-0.055556
3	-0.030474	-0.031250
4	-0.021391	-0.020000

Table 3: Hydrogen $(l = 0)$: $V(r) = -1/r$.

DISCUSSION

Across all three potentials, the imaginary-time propagation method reproduced the expected energy spectra with excellent accuracy. For the quartic potential, convergence was rapid: around 40,000 iterations sufficed to reach agreement with reference values to four significant figures. The PDMSE system also showed strong consistency, differing by less than one part in 10^3, despite the change of coordinate range from $x \in (-\infty,\infty)$ to $z \in [-\pi/2, \pi/2]$.

The hydrogen atom, however, presented a different numerical character. Its long-range 1/r potential and wide radial grid required nearly 200,000 iterations for comparable precision. Unlike the compact quartic or trigonometric wells, the Coulomb tail causes very gradual decay in imaginary time, so the ground-state projection proceeds much more slowly. Nevertheless, the energy values followed the correct inverse-square trend, confirming the validity of the approach.

Overall, the method's performance highlights the balance between numerical resolution, iteration count, and computational cost, finer grids and longer propagation yield higher accuracy but demand significantly more computation time.

CONCLUSION

The transformation to imaginary time provides a powerful way to reinterpret quantum mechanics in terms of relaxation rather than oscillation. By replacing $t \to -i\tau$, oscillatory terms $e^{-iEt/\hbar}$ become exponentially decaying $e^{-E\tau/\hbar}$, allowing wavefunctions to relax naturally toward the lowest-energy configurations. In this project, this principle was implemented numerically to extract successive eigenstates, where higher-energy components decay faster than the lower ones during imaginary-time propagation.

The same transformation also plays a fundamental role in theoretical physics, particularly in the Matsubara formalism [4], which describes quantum systems at finite temperature. There, the Wick rotation converts quantum evolution into a decaying behavior in imaginary time, allowing the system to be expressed in a thermally consistent way. In this sense, the same principle that allows numerical relaxation to a ground state here becomes essential in connecting quantum mechanics with statistical mechanics, forming a clear bridge between computation and theory.

[1] H. Ciftci, R. L. Hall, N. Saad. Asymptotic iteration method for eigenvalue problems. *J. Phys. A: Math. Gen.* **36**, 11807 (2003).

[2] S. Dong, *et al*. Semi-exact solutions to position-dependent mass Schrödinger problem with a class of hyperbolic potential V_0tanh(ax). *Eur. Phys. J. Plus* **131**, 176 (2016).

[3] S. H. Haider, Imaginary-Time Evolution in Quantum Systems – Python Codes, GitHub, 2025. https://github.com/syedhussainhaider137/imaginary-time-evolution.

[4] A. Altland and B. Simons, Condensed Matter Field Theory, 2nd ed. (Cambridge University Press, 2010).